

The Concept of pH :

The concept of pH, which stands for "potential of hydrogen," was first introduced by the Danish chemist Soren Peter Lauritz Sorensen in 1909. Working at the Carlsberg Laboratory, a research center established by the Carlsberg brewing company, Sorensen was tasked with studying the chemical processes involved in brewing, especially how acidity affected enzyme activity. In his experiments, he needed a more efficient way to measure the acidity or basicity of various solutions, which led him to the development of the pH scale.

Initially, Sorensen used a notation called "p[H]" to represent the concentration of hydrogen ions (H⁺) in a solution, with the "p" indicating a mathematical operation (logarithm) and "[H]" signifying the concentration of hydrogen ions. He proposed the pH scale as a logarithmic scale, where pH is defined as the negative logarithm (base 10) of the molar concentration of hydrogen ions in a solution. The formula is expressed as:

$$\text{pH} = -\log_{10}[\text{H}^+]$$

This means that for every unit decrease in pH, the concentration of hydrogen ions increases tenfold, making the solution more acidic.

History of pH Notation :

In 1909, the Danish chemist Soren Peter Lauritz Sorensen introduced the concept of pH at the Carlsberg Laboratory, originally using the notation "pH•", with H• as a subscript to the lowercase p. The concept was later revised in 1924 to the modern pH to accommodate definitions and measurements in terms of electrochemical cells. Around the early 1920s, during a period of continued refinement in chemical measurement techniques, Sorensen's original notation was simplified to just "pH", dropping the brackets and subscripts for easier use. The choice of the letter "p" has been debated over the years. Some suggest it stands for "power" (as in the French word "puissance"), or "potenz" in German, which Sorensen was familiar with. Others believe that "p" could stand for "potential" or could have been used simply as a mathematical symbol for indicating logarithmic calculation. There is even a theory that Sorensen

arbitrarily chose the letters "p" and "q" to distinguish between the test solution and a reference solution.

The modern definition of pH considers the activity of hydrogen ions rather than just their concentration. In chemistry, "activity" accounts for the effective concentration of ions, which is influenced by interactions with other ions in the solution. This approach provides a more accurate measurement in solutions with high ionic strength.

Early Methods of Measuring pH :

During the early 20th century, scientists developed several methods for measuring pH. Sorensen's original experiments involved using colorimetric indicators, which are substances that change color in response to different levels of acidity or basicity. For example, litmus, an organic dye extracted from lichen, turns red in acidic conditions and blue in basic conditions. This method was simple but not very precise.

The development of more sophisticated techniques came in the 1910s with researchers like Alice Catherine Evans and William Mansfield Clark. They focused on finding ways to measure the concentration of hydrogen ions more accurately in various solutions, especially for medical and biological applications. Clark's research on buffer solutions was especially important because buffers help maintain a stable pH in biological systems, which is critical for life processes.

The Invention of the pH Meter :

In 1934, a major advancement in pH measurement occurred when Arnold Orville Beckman, a professor at the California Institute of Technology, invented the first electronic pH meter. Beckman created the device initially to help Sunkist, a citrus grower, measure the acidity of lemons. His pH meter used a glass electrode sensitive to hydrogen ions and a reference electrode to measure the voltage difference, which corresponded to the pH of the solution. The invention was a breakthrough because it allowed for more accurate and consistent pH measurements than previous methods. Beckman later founded

Beckman Instruments, Inc., which became a leading company in the field of scientific instruments.

Understanding pH in Chemistry :

The pH level indicates how acidic or basic a substance is, determined by measuring the concentration of hydrogen ions in the solution. The pH scale, which ranges from 0 to 14, helps identify if a substance is an acid (0-6.9), neutral (7), or a base (7.1-14).

The pH Scale Explained :

- **0-6.9** : These values indicate an acidic substance, where 0 is the most acidic. For example, vinegar has a pH of 2.4.
- **7** : This is a neutral point, where substances like pure water fall.
- **7.1-14** : These values represent basic or alkaline substances, with 14 being the most alkaline.

How pH is Measured :

To find out a substance's pH level, indicators like litmus paper can be used. Litmus paper changes color when dipped in a solution, and the resulting color corresponds to a number on the pH scale, indicating if the substance is acidic, neutral, or basic.

Importance of pH Balance :

Maintaining a healthy pH balance in the body is crucial for overall well-being. The food and drinks you consume can affect your body's pH, and it's generally agreed that a balanced pH contributes to good health.

Scientific Explanation of pH :

The pH is calculated as the negative logarithm (base 10) of the hydrogen ion concentration in moles per liter. In simpler terms, the higher the concentration of hydrogen ions, the lower the pH (more acidic), and the lower the concentration, the higher the pH (more basic).

The pH scale itself is logarithmic, meaning each whole number change on the scale represents a tenfold change in acidity or alkalinity. For example, a substance with a pH of 3 is ten times more acidic than one with a pH of 4.

How pH is Measured Scientifically :

Standard methods to measure pH involve using a pH meter with a glass electrode or chemical indicators that change color according to the solution's acidity or alkalinity. Calibration is based on standard solutions whose pH values are internationally agreed upon.

Evolution of the pH Concept :

The concept of pH has evolved over time. Initially defined in 1909 by Sørensen as $p[H]$, the term "pH" came into broader use by 1924. In modern chemistry, $[H^+]$ represents the concentration of hydrogen ions. In dilute solutions, the pH calculation uses the ratio of $[H^+]$ to a standard concentration.

Measuring Hydrogen Ion Concentration Directly :

To measure hydrogen ions directly, electrodes calibrated for hydrogen ion concentration are used. A typical method involves titrating a strong acid with a strong base, using a background electrolyte to stabilize the reaction. The hydrogen ion concentration is then calculated and correlated with the measured potential.

pH vs. $p[H]$:

Though there is a slight difference between pH and $p[H]$ ($pH = p[H] + 0.04$), "pH" is commonly used to refer to both.

Understanding pOH :

The pOH measures the concentration of hydroxide ions (OH^-). It is calculated as the negative logarithm of the hydroxide ion concentration. The pOH can be derived from the pH value because the concentration of hydroxide and hydrogen ions are linked through the self-ionization constant of water (K_W). At room temperature, the relationship is:

$$pOH = 14 - pH$$

However, this equation may not hold in certain conditions, such as measuring soil alkalinity.

Average pH of common solutions		
Substance	pH range	Type
Battery acid	< 1	Acid
Gastric acid	1.0 – 1.5	
Orange juice	3.3 – 4.2	
Vinegar	4-5	
Black coffee	5 – 5.03	
Milk	6.5 – 6.8	Neutral
Pure water at 25 °C	7	
Sea water	7.5 – 8.4	
Ammonia	11.0 – 11.5	Base
Bleach	12.5	
1 M NaOH	14	

pH Indicators :

pH indicators are substances that change color when exposed to different pH levels, helping to determine whether a solution is acidic, neutral, or basic. The color change corresponds to a particular pH value, which can then be compared to a standard color chart to estimate the pH of the solution. This method provides a general pH range to the nearest whole number.

Universal Indicators :

A universal indicator is a combination of several pH indicators that together provide a continuous color change across a wide range of pH levels, usually from about pH 2 (acidic) to pH 10 (basic). This indicator is often used in the form of a universal indicator paper, which is absorbent paper treated with the universal indicator solution. When dipped in a liquid, the paper changes color according to the pH of the substance.

Measuring pH More Precisely :

For more accurate pH measurements, instruments like colorimeters or spectrophotometers can be used. These devices measure the intensity of the color change more precisely, giving a more detailed pH value compared to a standard color chart.

Electronic pH Meters :

Another reliable method for measuring pH is using an electronic pH meter. This device measures the voltage difference between a pH-sensitive electrode and a reference electrode, converting this difference into a pH reading. The pH meter provides a more precise and direct measurement of pH, making it useful for scientific and industrial applications where accuracy is important.

Unified absolute pH scale :

In 2010, a new approach to measuring pH was proposed, called the "unified absolute pH scale". This approach allows for a common reference standard to be used across different solutions, regardless of their pH range. The unified absolute pH scale is based on the absolute chemical potential of the proton, as defined by the Lewis acid–base theory. This scale is applicable to liquids, gases, and even solids. The advantages of the unified absolute pH scale include consistency, accuracy, and applicability to a wide range of sample types. It is precise and versatile because it serves as a common reference standard for pH measurements. However, implementation efforts, compatibility with existing data, complexity, and potential costs are some challenges.

Factors Affecting pH Measurements :

Accurate pH measurements can be influenced by various factors, especially at extreme values (very high or very low pH):

Electrode Limitations : The glass electrodes used in most pH meters can show errors at very high alkalinity (above pH 12) due to interference from sodium ions or potassium ions. This effect, known as "alkaline error," occurs because the glass electrode can also respond to these cations, not just hydrogen ions.

High Ionic Strength: In highly concentrated solutions, the activity of hydrogen ions can differ significantly from their concentration, leading to inaccurate pH readings. Special calibration techniques and electrodes are required to measure pH under these conditions.

Temperature Effects: The pH of a solution can change with temperature due to the shifting equilibrium between hydrogen ions and other species in the solution. Therefore, pH meters often have temperature compensation features to adjust for this effect.

pH in Different Contexts :

pH in Water :

In pure water at room temperature (25°C), the pH is 7, which means it is neutral. However, when water is exposed to the atmosphere, it can absorb carbon dioxide, forming carbonic acid, which lowers the pH to around 5.6, making it slightly acidic. This natural process occurs in rainwater, where the absorbed carbon dioxide makes the rain naturally acidic.

pH in Soil :

The pH of soil affects its ability to support plant life. Soils can range from acidic (pH 4-6) to alkaline (pH 8-9), depending on factors like rainfall, organic material, and the type of minerals present. Acidic soils may require lime to raise

the pH and make nutrients more available to plants, while alkaline soils may need sulfur to lower the pH.

pH in the Ocean :

Ocean pH is crucial for marine ecosystems. The ocean's surface water has an average pH of about 8.1, which is slightly alkaline. However, increased carbon dioxide levels in the atmosphere are causing more CO₂ to dissolve in seawater, forming carbonic acid and lowering the pH—a phenomenon known as ocean acidification. This affects marine life, particularly organisms that rely on calcium carbonate to form shells and skeletons.

Three pH Scales in Oceanography:

In oceanography, measuring pH is more complex than in pure water due to the unique chemical properties of seawater. This has led to the development of three distinct pH scales: the Free Scale (pHF), the Total Scale (pHT), and the Seawater Scale (pHSWS). Each scale accounts for different chemical species present in seawater, resulting in slight differences in pH values across these scales, often up to 0.10 pH units. The need for these different scales arises because traditional pH measurement methods, designed for solutions with lower ionic strength, are not suitable for seawater's higher ionic strength.

1. Free Scale (pHF) :

Definition : The free scale measures the concentration of free hydrogen ions ($[H^+]_F$) in seawater. It provides a straightforward representation of hydrogen ion concentration by focusing only on $[H^+]_F$ without accounting for interactions with other ions.

Limitation : Although this scale simplifies the representation of hydrogen ion concentration, estimating the free hydrogen ion concentration is challenging because the protonation constant of sulfate ions in seawater is difficult to measure accurately.

2. Total Scale (pHT) :

Definition : The total scale includes the free hydrogen ions ($[H^+]_F$) and the hydrogen sulfate ions ($[HSO_4^-]$), which form due to the protonation of sulfate ions in seawater. The equation representing the total concentration of hydrogen ions is:

$$[H^+]_T = [H^+]_F + [HSO_4^-]$$

Buffer Solutions : To accurately measure pH using the total scale, special buffer solutions with similar ionic strength to seawater are used. These buffers help resolve the differences in ionic strength that can affect electrode potential during pH measurement.

Practical Use : Because sulfate ions are abundant in seawater, the total scale is commonly used in oceanography for pH measurements.

3. Seawater Scale (pHSWS) :

Definition : The seawater scale accounts for the protonation of both sulfate and fluoride ions in seawater. It includes free hydrogen ions ($[H^+]_F$), hydrogen sulfate ions ($[HSO_4^-]$), and hydrogen fluoride ions ($[HF]$). The equation for the seawater scale is:

$$[H^+]_{SWS} = [H^+]_F + [HSO_4^-] + [HF]$$

Additional Complexity : Although the seawater scale considers interactions with both sulfate and fluoride ions, the practical difference between the seawater and total scales is minimal, since sulfate is far more abundant than fluoride in seawater.

pH in Food :

The pH of food influences its flavor, texture, and shelf life. Acidic foods, such as citrus fruits and vinegar, have a sour taste. Alkaline ingredients, like baking soda, can give baked goods a characteristic flavor and texture. Maintaining the

right pH is also important in food preservation; for instance, pickling involves lowering the pH of food to inhibit bacterial growth.

pH in the Human Body :

The human body maintains a stable pH balance, which is essential for health. Blood has a slightly basic pH of around 7.35 to 7.45, and even small deviations can be life-threatening. Different parts of the body have different pH levels; for example, the stomach has a very acidic pH (1.5-3.5) to aid digestion, while the skin's surface is slightly acidic (pH 4.5-6.5) to protect against harmful microbes.

pH and Chemical Reactions:

When calculating the pH of a solution, it's important to differentiate between strong and weak acids or bases:

Strong Acids and Bases : These substances dissociate completely in water. For example, hydrochloric acid (HCl) will fully ionize to release H^+ ions, while sodium hydroxide (NaOH) will fully release OH^- ions. The pH of a solution with a strong acid or base can be calculated directly from the concentration of the substance.

Weak Acids and Bases: These do not fully dissociate in water, meaning only a fraction of the molecules release H^+ or OH^- ions. This partial dissociation must be considered when calculating the pH. For instance, acetic acid (found in vinegar) is a weak acid, and ammonia is a weak base. Their pH values are determined using equilibrium expressions that account for the degree of dissociation.

Importance of pH :

1. Narrow pH Range for Living Organisms :

Living organisms can only thrive within a narrow range of pH levels. Even a small change in pH can have significant effects on their survival. For instance, acid rain has a pH level lower than 7, which makes the water more acidic. When

this acidic water flows into rivers, it lowers the overall pH of the river water, creating an environment that can be harmful or even lethal to aquatic life. Many fish and other organisms struggle to survive when the pH drops below their optimal range, leading to disruptions in ecosystems.

2. Digestion and Stomach Acidity :

Our stomachs contain hydrochloric acid (HCl), which is essential for breaking down food during digestion. However, when the stomach produces too much acid, particularly during episodes of indigestion, it can lead to discomfort and pain. To alleviate this discomfort, people often use antacids, which are mild bases. These antacids increase the pH in the stomach, neutralizing excess acid and helping to reduce irritation and pain.

3. Oral Health and Bacterial Activity :

The bacteria in our mouths can produce acids as they break down food particles, which can lower the pH of the oral environment. A lower pH can lead to tooth decay and other oral health issues. To combat this, it is recommended to use toothpaste, which is generally basic, to help restore a balanced pH in the mouth. Regular oral hygiene helps maintain an optimal pH level, preventing bacterial growth and protecting against cavities and gum disease.

4. Pain Relief from Bee Stings :

Bee stings introduce formic acid into the skin, which can cause considerable pain and irritation. To alleviate this discomfort, it is common to apply substances like baking soda, which is a mild base. This helps neutralize the acidic sting and maintain a balanced pH on the skin's surface, reducing pain and irritation from the sting.

In summary, maintaining proper pH levels is crucial for various biological processes, from digestion to oral health and beyond. Understanding the importance of pH can help us take better care of our health and the environment.